Predictors of radio-induced visual impairment after radiosurgery for uveal melanoma

Carmen Rosaria Gigliotti,[1] Giulio Modorati,[2] Maura Di Nicola,[2] Claudio Fiorino,[1] Lucia Alessia Perna,[1] Elisabetta Miserocchi,[2] Alberto Franzin,[3] Piero Picozzi,[3] Angelo Bolognesi,[4] Pietro Mortini,[3] Antonella del Vecchio,[1] Riccardo Calandrino[1]

[1]Department of Medical Physics, San Raffaele Scientific Institute, Milano, Italy
[2]Department of Ophthalmology, San Raffaele Scientific Institute, Milano, Italy
[3]Department of Neurosurgery, San Raffaele Scientific Institute, Milano, Italy
[4]Department of Radiotherapy, San Raffaele Scientific Institute, Milano, Italy

Correspondence to
Dr Antonella del Vecchio, Department of Medical Physics, San Raffaele Scientific Institute, Via Olgettina 60, 20132 Milano, Italy; delvecchio.antonella@hsr.it

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ABSTRACT

Aims The aim of the present work is to assess the main predictors of the most clinically relevant radio-induced effects after Gamma Knife stereotactic radiosurgery (GKRS) for uveal melanoma (UM).

Materials and methods Medical records and three-dimensional dosimetry data of critical structures of 66 patients were retrospectively reviewed. Cox's proportional hazard model was used to identify clinical and dosimetric variables as independent risk factor for GKRS-related complications.

Results The fraction of the posterior segment receiving more than 20Gy (V20), Bruch's membrane rupture and tumour thickness were significant prognostic factors for neovascular glaucoma. A clear relationship with the dose received by 1% of the optic nerve (D1%) was found for radiation retinopathy and papillopathy. Multivariables models resulted for visual acuity (VA) reduction >20% of the basal value and for complete VA loss, both including largest tumour diameter and D1% to the optic nerve. The predictive model for complete VA loss includes also Bruch's membrane rupture. An alternative model for complete visual acuity loss, including the optic nerve-prescription isodose minimum distance, was also suggested.

Conclusions We found clinical and dosimetric variables to clearly predict the risk of the main side effects after GKRS for UM. These results may provide dose constraints to critical structures, potentially able to reduce side effects. Constraining D1% to the optic nerve below 12-13Gy may result in a dramatic reduction of blindness risk, while reducing V20 of the posterior segment of the bulb could limit the neovascular glaucoma onset.

INTRODUCTION

Uveal melanoma (UM) represents the most common primary intraocular malignancy in adults, with 2.8 cases/million/year in Europe.[1] The quoad vitam prognosis of this disease is discouraging since it is a life-threatening tumour with an overall mortality rate of up to 40%–50% in 15 years, mainly due to liver metastasis.[2] Enucleation has been the traditional treatment for UM, although, since it is not advantageous regarding metastatic spread and survival,[3] conservative management of the malignancy using ionising radiation is preferred when possible. Different radiation delivery techniques are used for the UM treatment including plaque brachytherapy,[4] proton therapy,[5] stereotactic radiotherapy[6 7] and stereotactic radiosurgery. Gamma Knife stereotactic radiosurgery (GKRS) is a well-assessed strategy for conservative treatment of UM providing satisfactory results in terms of survival, local tumour control and eye preservation.[8 9] A wide range of prescription radiation doses (from 20 to 80Gy) was used in published series of GKRS for UMs,[10–12] showing a de-escalation trend over the years with the aim of reducing side effects.[13] Actually, several complications following GKRS have been reported, with a complication rate ranging from 55% to 82% in the different studies.[14 15] Furthermore, deterioration of visual acuity consequent to the treatment is an adverse event that significantly reduces patients' quality of life.[16] An adequate planning optimisation, in terms of definition of the prescribed dose and dose limitations to critical structures, may potentially reduce the incidence of severe side effects. Nevertheless, despite side effects that are frequently observed, literature studies designed to investigate dose–effect relationship of critical structures in this specific situation are lacking. The purpose of the present study is to find dosimetry and clinical parameters associated with an increased risk of the main radiation-induced effects in patients affected by UM treated with exclusive GKRS.

MATERIALS AND METHODS

Recovering dosimetry and clinical data

From 1994 through 2014, 149 consecutive patients were treated with exclusive GKRS for UM at San Raffaele Hospital (Milan, Italy). In particular, the Elekta Leksell Gamma Knife model C was used until 2007, later replaced by a Perfexion. Before treatment, patients underwent high-resolution T1 and T2 MRI with gadolinium. Tumour borders were recognised on MRI sections in three-dimensional reconstructions and drawn, including an additional 2 mm margin around melanoma edges in the lateral direction. The dose distribution was normalised to the maximum dose, while the tumour periphery was always included in the 50% isodose curve. To improve dosimetric accuracy, skull dimension was defined on the CT image: a tissue equivalent gel bolus was used for superficial tumours (ie, less than 1 cm from the external body contour) in order to guarantee charged-particle equilibrium. Collimator size and isocentres were chosen in order to strictly tailor the dose distribution to the tumour shape and to spare the remaining structures of the eye in order to reduce the risk of side effects. Nevertheless, no critical structures were routinely contoured during

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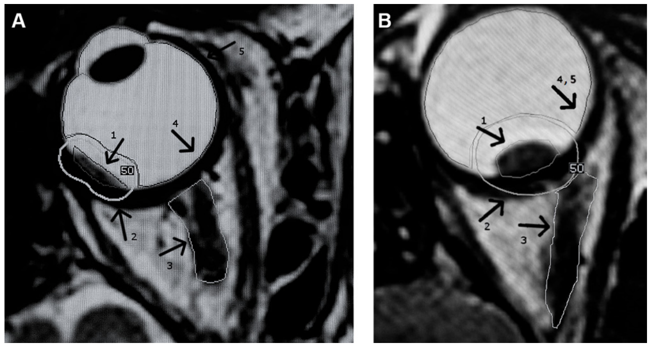

Figure 1 The image shows two examples (tumour distant in (A) and close to the optic nerve in (B) of the critical structures contoured on the MRI used for treatment planning. Arrow n°1 points the target volume (melanoma), n°2 points the 50% prescription isodose, n°3 the optic nerve, the posterior segment of the globe is pointed by arrow n°4 and arrow n° 5 points the eyeball, which includes the posterior segment and the lens.

treatment planning except for selected cases in which we tried not to exceed 12Gy to the optic nerve. The treatment was performed in a single fraction of 30 to 50Gy at the 50%isodose after a careful immobilisation of the globe. As previously described by Modorati *et al*,[9] patients received retrobulbar anaesthesia with long-acting agents (5cm^3 of 1% ropivacaine) in order to obtain complete akinesia, after which two extraocular muscles, chosen according to tumour location to optimise globe position for the treatment, were sutured through the conjunctiva using 3.0 black silk suture. The threads were then fixed to the stereotactic frame to immobilise and orientate the globe with the tumour as closely as possible to the centre of the stereotactic frame.

The prospectively filled medical records of all patients were recovered. Three-dimensional dosimetry data of the involved critical structures could be analysed for patients treated from 2007 (66/149). In particular, optic nerve (ON), the healthy volume of the eyeball and of the posterior segment of the globe (ie, excluding the volume covered by the prescription isodose) were recontoured on MRI sections by a single observer (figure 1) and dose–volume histograms (DVHs) were calculated. Target volume, prescription dose, number of isocentres and the minimum distance between ON and prescription isodose were also recovered. The following clinical data were recorded before treatment: basal visual acuity (VA), tumour thickness, largest tumour diameter (LTD), Bruch's membrane rupture (present or absent), tumour location (anterior or posterior to equator), quadrant involved by tumour and minimum distance from ON. All patients underwent complete ophthalmological examination before treatment, and then at 1 day and 1, 3 and 6 months after GKRS. The clinical follow-up and VA was prospectively updated every 6 months. For the 66 patients enrolled in the study, the median follow-up was 24 months (range 5–67 months).

Definition of end points

The following GKRS-related complications were analysed: radiation maculopathy (RM), radiation retinopathy (RR), radiation papillopathy (RP), optic nerve atrophy (ONA), neovascular glaucoma (NVG), exudative retinal detachment (RD) and phthisis. The presence of these complications was clinically evaluated by slit lamp examination and indirect ophthalmoscopy of the fundus oculi. Ancillary testing including ultrasound, optical coherence tomography and fluorescein angiography was performed when needed. In addition, a reduction of VA greater than 20% of the basal value ($\Delta VA \geq 20\%$) and the complete loss of the basal VA ($\Delta VA = 100\%$) were considered as end points. The presence of cataract was clinically evaluated, but intentionally excluded from the analysis in order to focus on pathology causing irreversible deterioration of visual function.

Analyses

The actuarial risk for the end points was estimated using Kaplan-Meyer method. Univariate Cox's proportional hazard model was used to identify clinical and dosimetric variables associated to an increased risk of complications. To assess the dose–volume parameters better discriminating between patients with and without side effects, the two populations' DVHs were compared using two-sided t-tests; the procedure was applied to the three considered structures for the end points with a number of events >6. The most significant DVH parameters ($p<0.05$, t-test), selected in correspondence of the minimum p value, were included in Cox's analyses.

Finally, variables with $p<0.10$ at the univariate analysis were entered into a backward multivariable Cox's regression, keeping in the resulting model variables with $p<0.05$. For all significantly associated variables, receiver operating characteristic (ROC) curve analyses were used to assess most predictive cut-off values, through the corrected Youden index. The discriminative power of univariate and multivariate models was measured by area under curve (AUC).

Furthermore, the end points concerning VA reduction were evaluated at the median follow-up and a logistic regression was performed to evaluate the influence of the tumour-ON distance.

Analyses were performed using MedCalc statistical software (Mariakerke, Belgium).

RESULTS

Patient population

Table 1 summarises patients' characteristics. Median age at the time of GKRS was 68 years. In 78.8% of cases, the tumour was located in the posterior to equator position. Median prescription dose was 35Gy (range 27–35Gy) in single fraction, prescribed to the 50% isodose for the majority of the population (median 50%, range 48%–60%). Median target volume was 0.35 cm^3. With the exception of one patient, for whom enucleation was necessary, local tumour control was always obtained. Five patients (7.6%) developed liver metastasis. None of the patients died during follow-up. Seven patients (11%) experienced NVG, 11 (17%) RR, 11 (17%) RP, 6 (9%) RM, 1 (2%) ONA and 20 (30%) cataract. Sixteen patients (24%) showed exudative RD at the time of diagnosis and six (9%) experienced it after GK. During follow-up, exudative RD resolved in two of our patients (after 25 months on average). No patients experienced phthisis. Moreover, five patients(7.6%) had improved, 19 (28.7%) had stable and 42 (63.6%) had a worse VA after GKRS. In particular, 42 patients experienced a reduction greater than 20% of VA and 15 completely lost VA.

A threshold of 10% incidence was established to include toxicity in statistical analyses; therefore, due to absence or extremely low incidence of the events, exudative RD, RM, ONA and phthisis were excluded.

Analyses for the risk of RR, RP and NVG

The 2 years risk was 19% for RR, 12% for RP and 14% for NVG (figure 2). In table 3, a summary of univariate Cox's analyses is reported, showing clinical and dosimetric variables most predictive of significant risk of adverse events. A clear relationship

Table 1 Patients, tumour and treatment parameters

Age		
Mean (median)	64.9 years	(68 years)
Range	42–84 years	
Sex	N	%
Male	38	57.6
Female	28	42.4
Basal VA value (Snellen notation)		
Mean (median)	0.46	(0.4)
Range	(0–1.0)	
Thickness		
Mean (median)	6.5 mm	(6.1 mm)
Range	2.6–11.7 mm	
Largest tumour diameter		
Mean (median)	11.0 mm	(11.4 mm)
Range	5.3–18.5 mm	
Bruch's membrane rupture	N	%
Yes	21	31.8
No	45	68.2
Position	N	%
Posterior to equator	52	78.8
Anterior to equator	14	21.2
Quadrant	N	%
Posterior pole	9	13.6
Inferior	26	39.4
Temporal	13	19.7
Superior	11	16.7
Nasal	7	10.6
Prescription isodose–optic nerve distance		
Mean (median)	3.79 mm	(3.05 mm)
Range	0–13.9 mm	
Prescription dose		
Mean (median)	33 Gy	(35 Gy)
Range	27–35 Gy	
Target volume		
Mean (median)	0.49 cm^3	(0.35 cm^3)
Range	0.07–2.06 cm^3	
N° shots		
Mean (median)	5	(5)
Range	1–17	

The incidence of each side effect is reported in table 2.
VA, visual acuity.

with maximum dose to the ON, defined as the dose to 1% of the volume, was found for RR (p=0.0004) and for RP (p=0.0099). The best cut-off value resulted D1%(ON) >14.9 Gy, meaning that 14.9 Gy better discriminates patients affected by RR or RP from those who do not develop these side effects. As expected, the greater the distance of ON from prescription isodose, the lower the probability to develop retinopathy. As a matter of fact, prescription isodose-ON minimum distance resulted a protective variable (p=0.02) with a HR=0.56 (per mm). Similarly, the anterior to equator position resulted a protective variable (p=0.009) with HR=0.2.

Concerning NVG, the volume of posterior segment receiving more than 20 Gy (p=0.001,cut-off: V20 >413.7 mm^3), tumour thickness (p=0.0008, cut-off: thickness >8.7 mm) and Bruch's membrane rupture (p=0.0064) were the most predictive factors.

Models for the risk of VA deterioration

The 2 years risk for ΔVA ≥20% and for ΔVA=100% was 61% and 27%, respectively (figure 2). Multivariate Cox's analyses resulted in a two-variables predictive model for ΔVA ≥20% (AUC=0.72) and in a three-variable model for ΔVA=100% (AUC=0.84) in which LTD and D1%(ON) were associated with the risk of VA deterioration. The best cut-off values for LTD were 11.6 mm for ΔVA ≥20% and 9 mm for ΔVA=100%; the best cut-off values for D1%(ON) were 8.5 Gy for ΔVA ≥20% and 13.2 Gy for ΔVA=100%. Bruch's membrane rupture was the third predicting factor for ΔVA=100% risk. Concerning blindness (ΔVA=100%), from the analyses a second model, with a similar discriminative power (AUC=0.87), resulted, including the minimum distance of ON from the prescription isodose as the most significant dosimetric prognostic factor (p=0.005, HR=0.54). Table 4 summarises the results of multivariable analyses.

figure 3 shows actuarial risk curves of ΔVA ≥20% splitting the population in two groups: patients that received D1%(ON) below the cut-off value and/or have LTD smaller than the cut-off value against patients that present both parameters above the cut-off values. The difference of the two actuarial curves resulted significantly with p=0.008. The same procedure was applied to VA=100% for both the found models. In both cases, the risk curves differed significantly (p<0.0001). figure 4 shows Kaplan-Mayer curves for the two grouping.

Furthermore, we developed a risk probability map for VA deterioration as a function of the minimum distance of tumour from ON (figure 5). The end points were evaluated at the median follow-up (2 years); thus, the cohort is reduced to n=54 (38 events) for ΔVA ≥20% and n=44 (14 events) for blindness. From univariate analysis, this variable resulted as a significant predictive factor for ΔVA ≥20% with p=0.005 (OR=0.77 95% CI 0.64 to 0.93) and for blindness with p=0.05 (OR=0.74 95% CI 0.55 to 1). The AUC of the models are respectively 0.74 (95% CI 0.6 to 0.85) and 0.71 (95% CI 0.56 to 0.84).

DISCUSSION

The primary goal of UM conservative treatment is patient survival with local tumour control. A second main objective is eye preservation and sparing of visual function. Several studies demonstrated that GKRS is a treatment modality to satisfactorily achieve both results.[8–12] The optic pathways are particularly radiosensitive and it is not surprising that high doses delivered in UM radiosurgery treatment are associated with development of rather severe side effects. The evidence of the insurgence of complications after GKRS treatment for UM is well known from literature,[14–16] but the relationship with dose–volume parameters of structures involved in visual functionality has not yet been

Table 2 Side effects incidence and onset time

Side effect	Incidence (%)	Median onset time (range) (months)
Visual acuity reduction ≥20%	42/66 (63.6%)	6 (6–24)
100% Visual acuity reduction	15/66 (22.7%)	6 (6–36)
Radiation retinopathy	11/66 (17%)	19.5 (12–41.9)
Neovascular glaucoma	7/66 (11%)	15.6 (11.8–37.1)
Radiation papillopathy	7/66 (11%)	19.5 (12–41.9)
Radiation maculopathy	6/66 (9%)	19.3 (10.2–43.7)
Exudative retinal detachment	6/66 (9%)	7.9 (2.3–31.1)
Optic nerve atrophy	1/66 (2%)	37.5
Cataract	20/66 (30%)	10.2 (3.3–22.4)
Phthisis	0/66 (0%)	

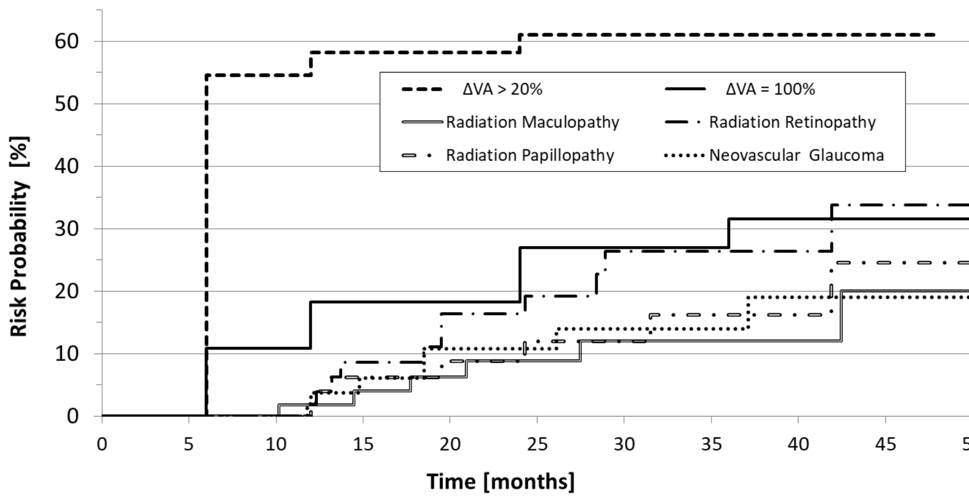

Figure 2 Kaplan-Mayer actuarial risk curve of the main radio-induced pathologies experienced in the cohort of patients considered in our analyses. VA, visual acuity.

clearly modelled for radiosurgery, limiting de facto the potential of radiosurgical planning to reduce these events. This work focused on the assessment of quantitative correlation between DVHs of involved critical structures and clinically reported toxicity. Selected clinical variables were also included in the analyses. The findings of this investigation, despite the relatively small number of patients, showed the existence of clear quantitative correlations for several end points.

Table 3 Univariate Cox's analysis of factors predicting the main radio-induced side effects

Predictive factor/ end point	Neovascular glaucoma	Radiation retinopathy	Radiation papillopathy
D1% optic nerve		p=0.0004; HR=1.5; 95% CI 1.07 to 1.24; cut-off: 14.9Gy; AUC=0.85 95% CI 0.74 to 0.92	p=0.0099; HR=1.14; 95% CI 1.03 to 1.26; cut-off: 14.9 Gy; AUC=0.83 95% CI 0.72 to 0.91
V20 posterior segment	p=0.001; HR=1.003; 95% CI 1.001 to 1.005; cut-off: 413.7 mm^3; AUC=0.83 95% CI 0.71 to 0.91		
Prescription isodose–optic nerve distance		p=0.022; HR=0.56; 95% CI 0.34 to 0.92; cut-off: 2.2 mm; AUC=0.81 95% CI 0.69 to 0.89	
Position: anterior to equator		p=0.009 HR=0.2; 95% CI 0.06 to 0.66	
Tumour thickness	p=0.0008; HR=2.01; 95% CI 1.34 to 3.02; cut-off: 8.7 mm; AUC=0.83 95% CI 0.72 to 0.91		
Bruch's membrane rupture	p=0.0064 HR=10.11; 95% CI 1.93 to 53.00		

AUC, area under ROC curve.

Tumour thickness was the clinical factor more strongly associated with the risk of NVG, together with Bruch's membrane rupture and volume of the posterior segment receiving more than 20 Gy. This can be explained by radiation capability of damaging blood vessels that could cause retinal ischaemia inducing the secondary glaucoma.[17] The maximum dose received by ON was found to be a strong predictor for RR and RP, with a strongly discriminative threshold: 14.9 Gy.

The visual impairment, defined according to VA, associated to the radiation-induced optic neuropathy (RION) was studied to evaluate the optic apparatus tolerance to radiation for single-fraction radiosurgery. After an early conservative constraint of a maximum dose of <8 Gy for the optic apparatus tolerance,[18] different groups subsequently reported low risk for vision loss with higher single-fraction doses. Actually, Laber *et al*[19] reported a significant risk for RION with a D1% ≥10 Gy. Stafford *et al*[20] and, similarly, Pollock *et al*[21] observed a very low risk of RION using an optic structure constraint of 12 Gy. Also, the more recent literature[22] indicates a low risk of RION treating perioptic tumours with single-fraction or hypofractionated stereotactic radiosurgery with a maximum dose to optic pathway of 12 Gy in one, 19.5 Gy in three and 25 Gy in five fractions. Furthermore, Polishchuck *et al*[23]

Table 4 Multivariate Cox's analysis of factors predicting the deterioration of the basal visual acuity

Predictive factor/end point	20% Visual acuity reduction (Model AUC=0.72; 95% CI 0.59 to 0.82)	100% Visual acuity reduction (Model AUC=0.84; 95% CI 0.73 to 0.92)	100% Visual acuity reduction (Model AUC=0.87; 95% CI 0.77 to 0.94)
D1% optic nerve	p=0.05; HR=1.04 95% CI 1.00 to 1.08; cut-off: 8.5 Gy	p=0.001; HR=1.14 95% CI 1.06 to 1.24; cut-off: 13.2 Gy	
Prescription isodose–optic nerve distance			p=0.005; HR=0.51; 95% CI 0.32 to 0.82; cut-off: 3.9 mm
Largest tumour diameter	p=0.03; HR=1.14 95% CI 1.01 to 1.28; cut-off 11.6 mm	p=0.01; HR=1.41; 95% CI 1.08 to 1.83; cut-off: 9.0 mm	p=0.0041; HR=1.55; 95% CI 1.15 to 2.08; cut-off: 9.0 mm
Bruch's membrane rupture		p=0.0042; HR=5.12; 95% CI 1.68 to 15.59	p=0.0083; HR=4.40; 95% CI 1.47 to 13.

AUC, area under ROC curve.

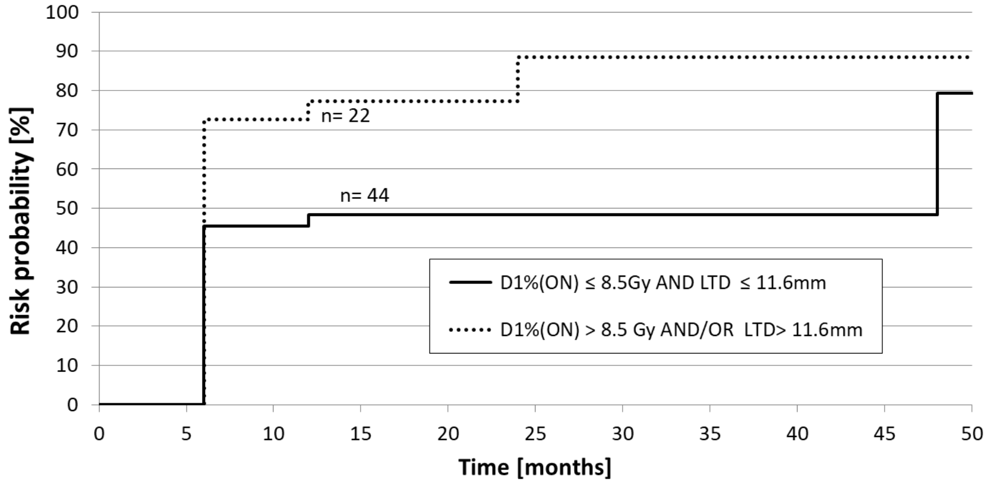

Figure 3 Kaplan-Mayer actuarial incidence curve of ΔVA ≥20% splitting the population in two groups: the patients that received a maximum dose to the optic nerve below the cut-off value (8.5 Gy) and/or have the largest tumour diameter (LTD) smaller than the cut-off value (11.6 mm), with solid line, against the patients that present both parameters above these values, with dashed line.

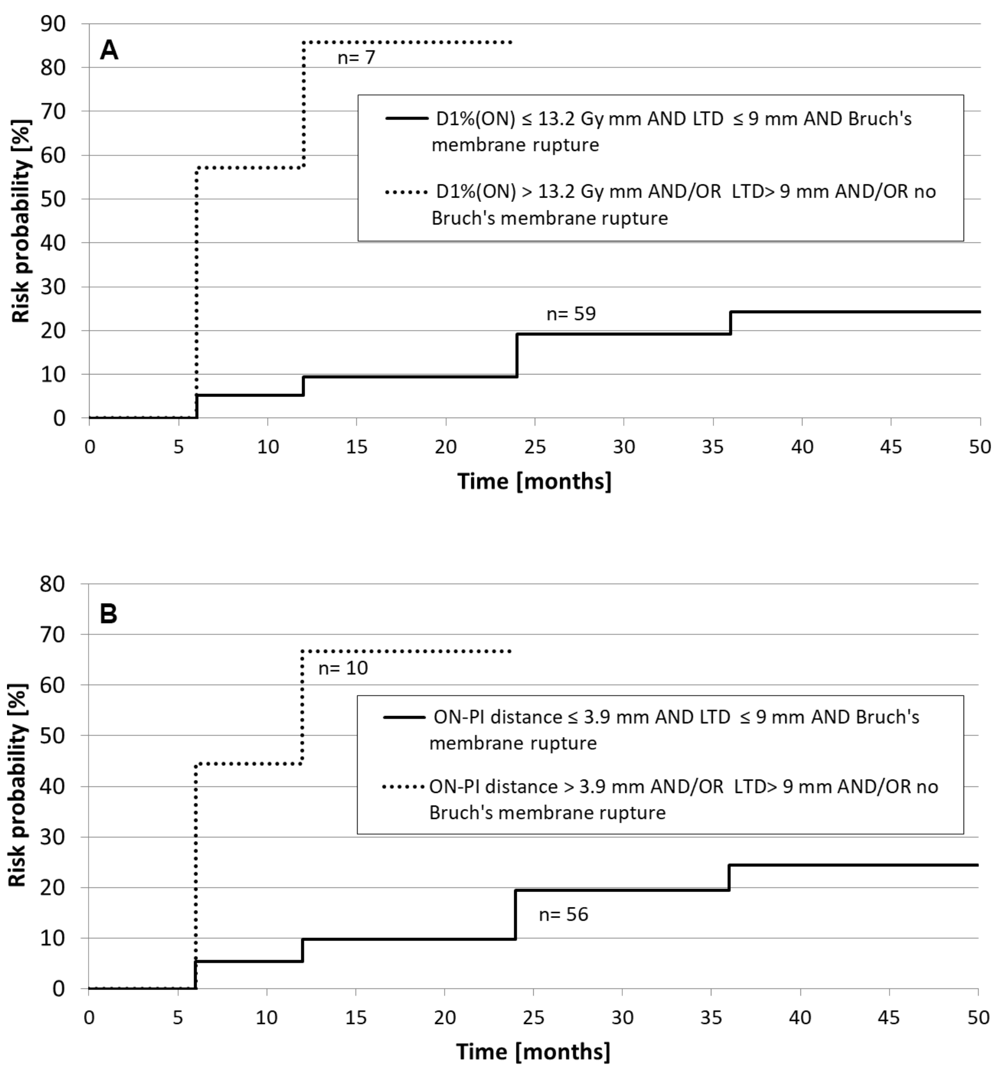

Figure 4 Kaplan-Mayer actuarial incidence curve of ΔVA ≥100% splitting the population in two groups: the patients that have the values of the predictors above the cut-off values (dashed lines) and the patients that have at least one of the predictor under the cut-off values (solid lines). (A) Refers to the model including D1% to the optic nerve (ON), largest tumour diameter (LTD) and Bruch's membrane rupture and (B) refers to the model including the ON prescription isodose minimum distance largest tumour diameter and Bruch's membrane rupture.

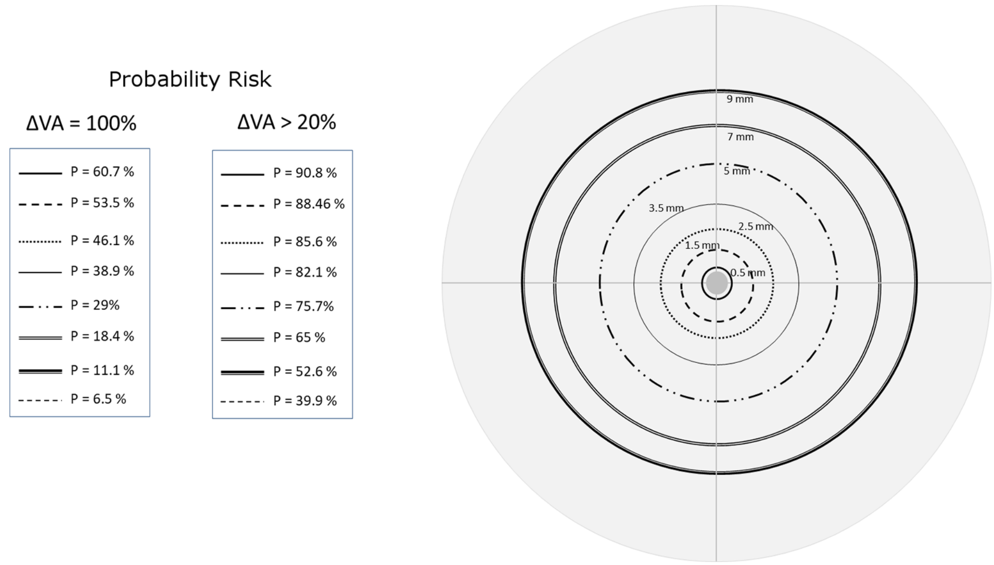

Figure 5 Schematic representation of the eye with isodistance lines form the optic nerve (central grey circle). In the box, the probability risk of visual acuity (VA) reduction greater than 20% and blindness at 2years for a tumour located at the corresponding distance from the optic nerve.

show a strong correlation between visual outcome and V28GyE of the macula and ON for the UM patients treated with proton beam.

In our study, we found that volumetric constraints to ON are less significant than the maximum dose as toxicity predictor. Actually, the suggested multivariable predictive models both for VA reduction (ΔVA ≥20%) and for complete blindness (ΔVA=100%) include the D1% to ON as dosimetry variable.

Concerning ΔVA >20%, we observed that in the early reduction (from 6 months to 3 years), the probability risk is about one-half for the population that have predictive factors not exceeding cut-off values (figure 4b), whereas this population has a probability risk for late reduction comparable with the population that exceeds the cut-off values. In fact, the 4 years risk differs only by 10%, suggesting that keeping D1%(ON) <8.5 Gy could be effective only in postponing the onset of VA reduction.

Furthermore, the dose to ON should not exceed the threshold of 13.2 Gy to reduce the risk of complete VA loss. In this case, tumour physical dimension and Bruch's membrane rupture are the most significant clinical risk factors. The actuarial incidence for the population having these prognostic factors shows a clear difference as compared with the other patients (figure 4a). In particular, the 1-year risk is about seven times greater, switching from 10% to 67% probability. The found ON radiation tolerance for vision loss of 13 Gy is in line with the values reported in literature and with the QUANTEC (Quantitative Analysis of Normal Tissue Effects in the Clinic) guidelines.[24]

Interestingly, the predictive model for blindness risk including minimum distance of ON from prescription isodose as dosimetric prognostic factor underlines the influence of dose conformation to the target on GKRS-related complications. Furthermore, this model provides a constraint independent from prescribed dose and easily measurable during treatment planning.

CONCLUSIONS

We found clinical and dosimetric variables to clearly predict the risk of the main side effects after GKRS for UM. These results may provide dose constraints to critical structures, potentially able to reduce side effects. Reducing the V20 of the posterior segment of the bulb could limit the NVG risk, while constraining D1% to ON below 12–13Gy may result in a dramatic reduction of blindness risk.

The distance of the tumour from the ON can be seen as a surrogate of the dose received from the latter and could be used to predict VA deterioration. The plot of the probability risk as a function of this variable could be useful, once the model is validated by expanding the cohort of patients, as a practical tool for clinicians for a preliminary treatment evaluation.

Contributors CR, dVA and MG conceived of the study. MG, ME, DNM, FA, PP, BA and MP initiated the study design including patient recruitment and clinical data collection. GCR, dVA and PLA conducted the dosimetric study including data collection and statistical analysis. FC provided statistical expertise in clinical trial design and GCR is conducting the primary statistical analysis. GCR prepared the manuscript draft. All authors contributed to refinement of the study protocol and approved the final manuscript.

Competing interests The corresponding author AdV has a contract as starter-up with Elekta Instrument AB. No other conflict of interests to declare for the other authors.

Patient consent Obtained.

Ethics approval IRCCS Ospedale San Raffaele Ethics Committee.

Provenance and peer review Not commissioned; externally peer reviewed.

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